UMassAmherst
College of Engineering
Biomedical Engineering

Hunger Qs
Using Electrophysiology to Measure Hunger Signals in the Stomach

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Motivation

- Individuals with eating disorders can miss hunger cues due to interoception feelings like stress
- **Goal**: help individuals recognize true hunger cues in order to improve nutritional intake and overall health via electrogastrogram (EGG) device

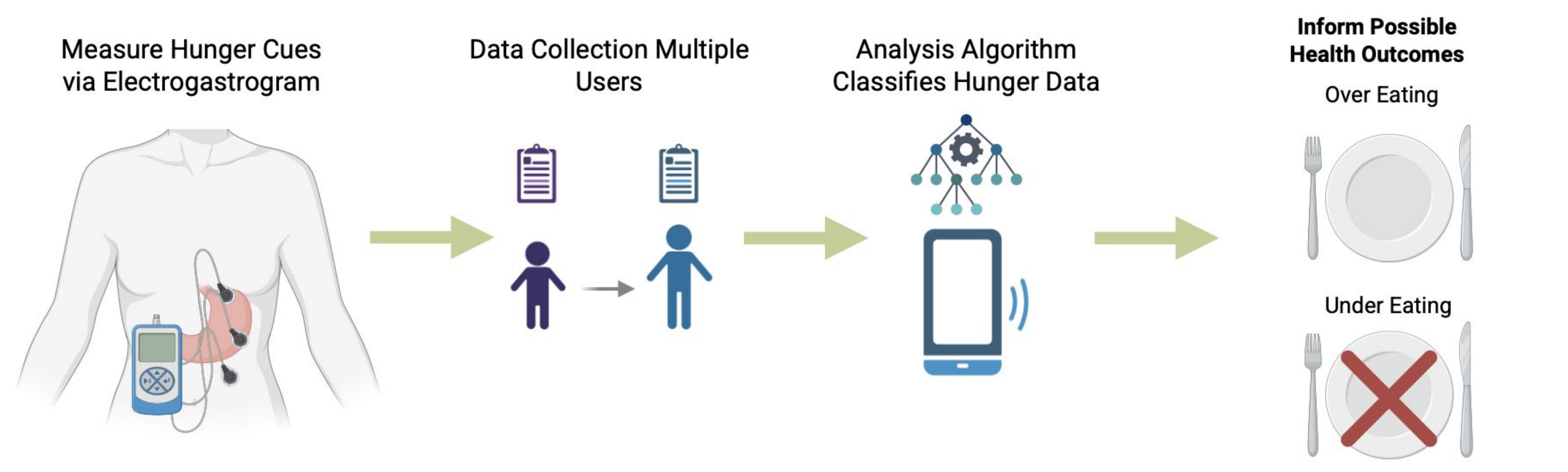
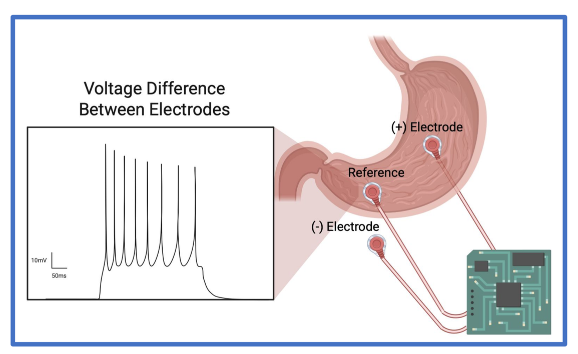


Figure 1. Graphical abstract detailing the methodology and outcomes of proposed hunger sensing device.

Background: Commercial EGG System (iWorx)



Gastric contraction frequency: 0.015 - 0.15Hz [1].

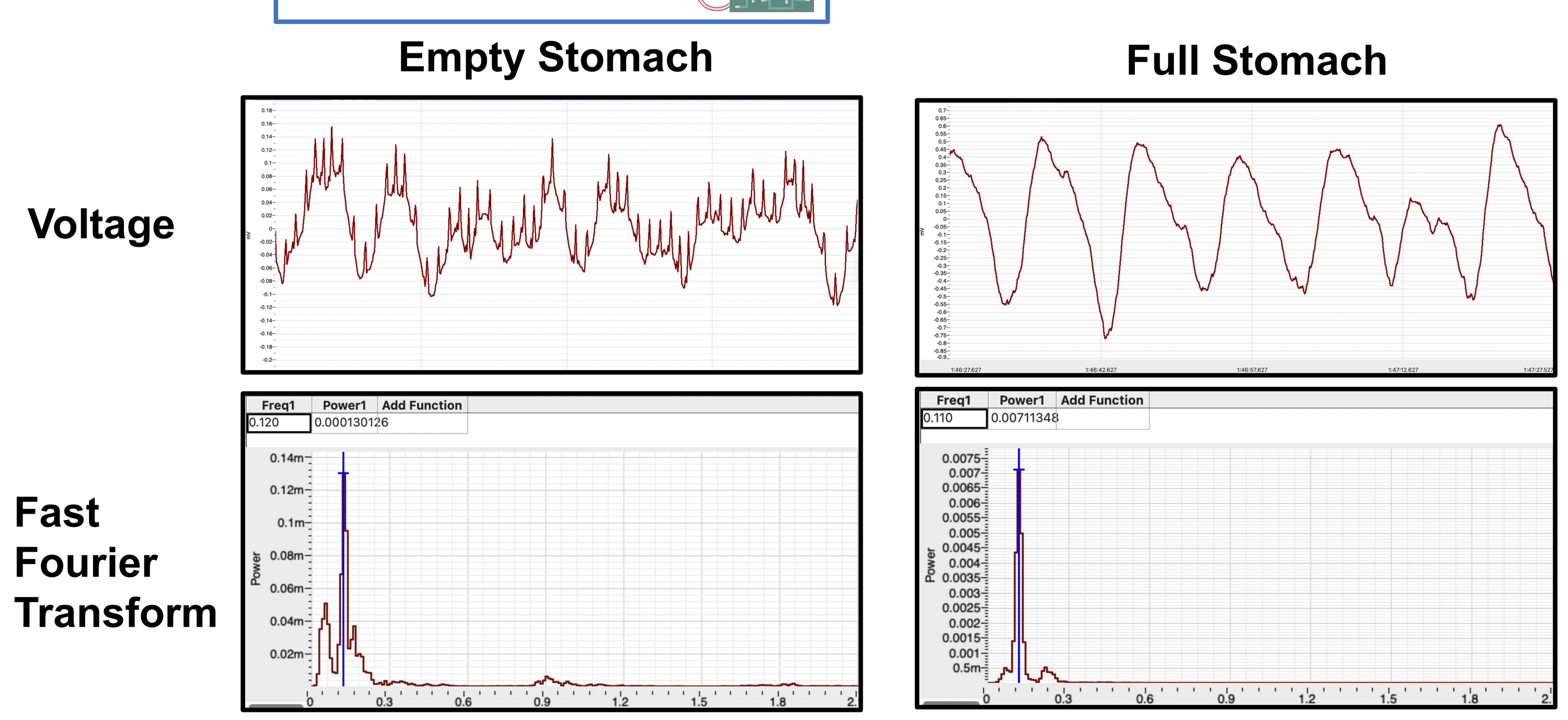


Figure 2. EGG measurements on empty and full stomach using iWorx. Dominant frequencies fell within gastric domain, with higher peak power in full condition.

α Prototype: From-Scratch EGG Circuitry to Sense Hunger

Step I: Amplify low frequency gastric wave signal read from electrodes.

Step II: Remove noise from other voluntary (ie. abdominus rectus) and involuntary (ie. diaphragm) muscles.

I. Pre-amplification Circuit

II. 4th Order Butterworth Filter

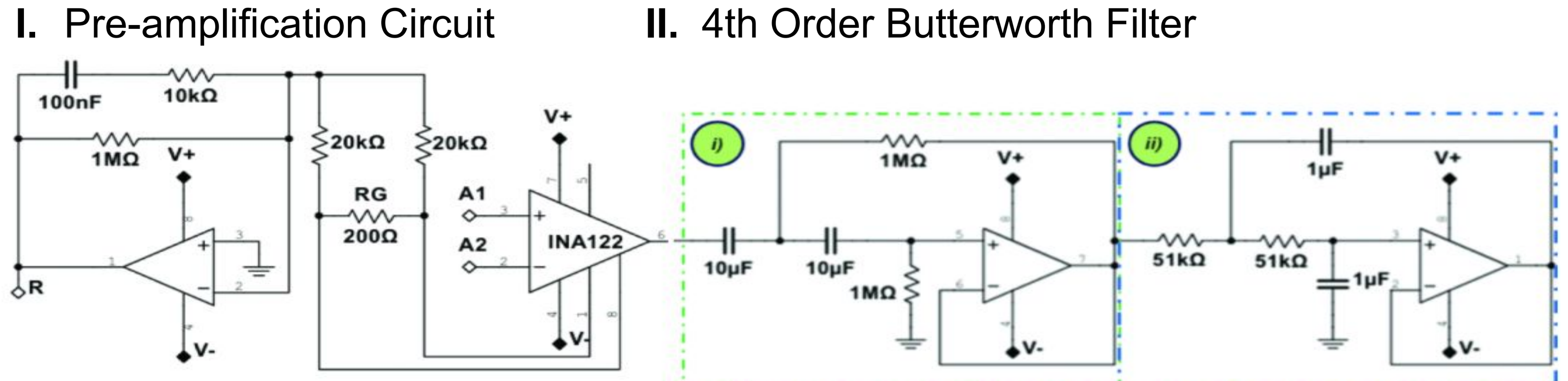


Figure 3. Fourth order butterworth filter for EGG inspired by Ochoa et al. [2].

β Prototype: From-Scratch EGG Ciruit and Algorithm Development

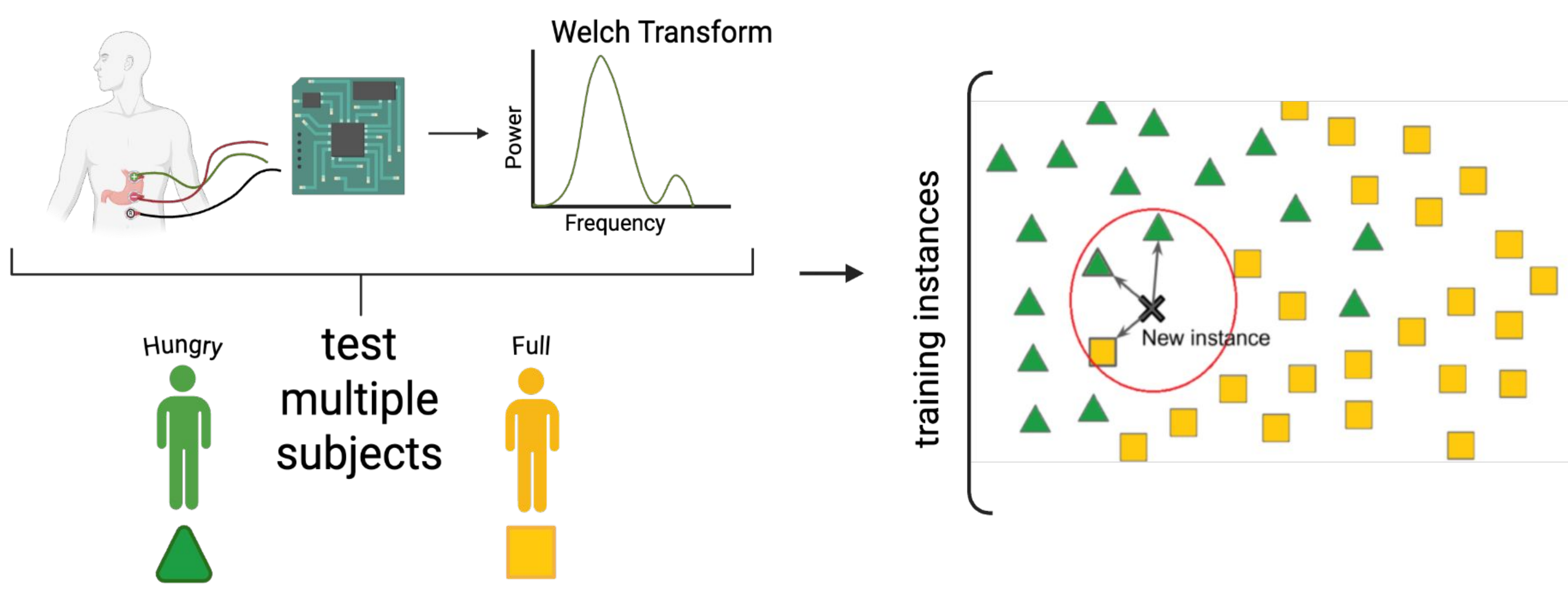


Figure 4. Data collection and processing using from-scratch EGG, Welch analysis algorithm and K-Nearest Neighbors (kNN) machine learning model.

Validation and Verification of Design

A. Accuracy and Noise Removal Ability of Welch Function Testing

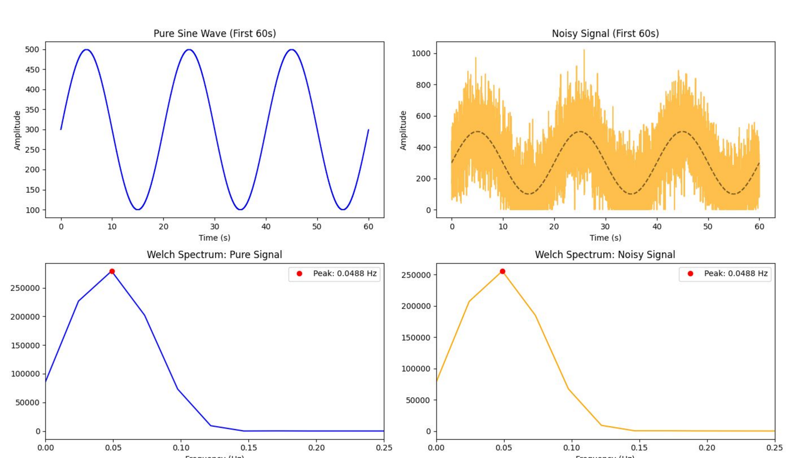


Figure 5. Welch analysis on pure and noisy sine waves function in capturing the dominant frequency.

Condition	Empty			Full		
Subject	1	2	3	1	2	3
FFT Lying	0.06	0.12	0.06	0.16	0.10	0.10
Welch Lying	0.06348	0.12501	0.04883	0.06348	0.08789	0.10254
Welch Sitting	0.04883	0.13672	0.05371	0.05371	0.10742	0.12695
Welch Walking	0.04295	0.1416	0.05372	0.06348	0.122207	0.20508

Table 1. Comparison in peak frequency (Hz) analyzed by FFT and Welch on empty and full stomach conditions in different postures in three subjects.

B. Machine Learning (kNN) Accuracy Testing

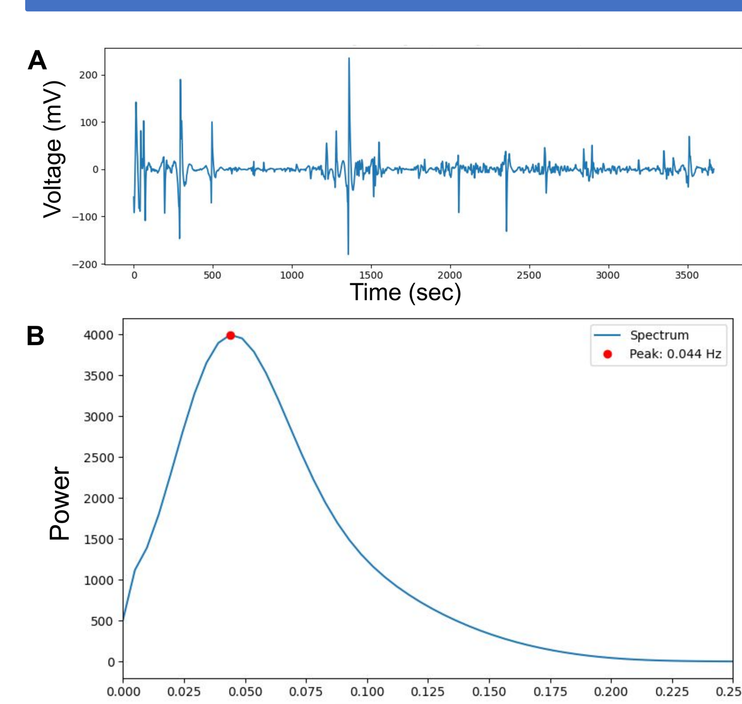
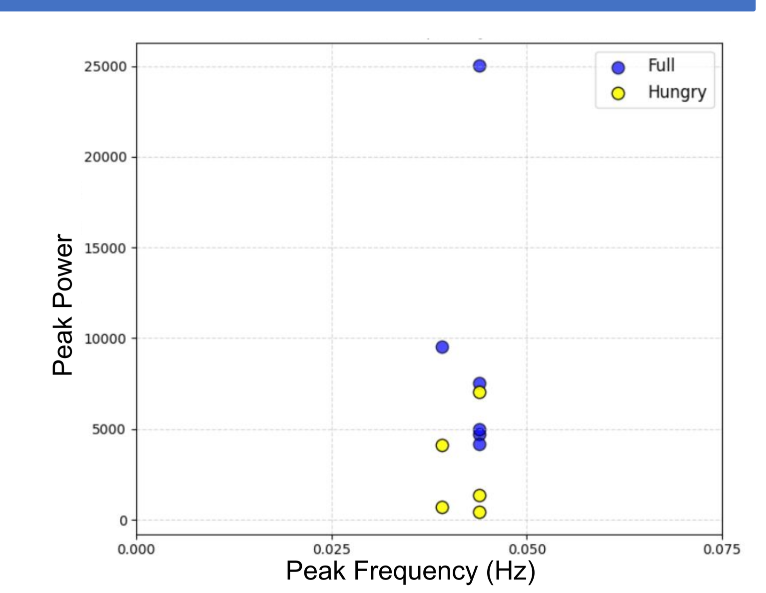


Figure 6. (A) Raw EGG spectrum, **(B)** Dominant peak frequency analysis using Welch function.



Accuracy = 77% (n = 100)

Figure 7. Scatter plot between dominant frequency and peak power showing clusters for kNN classification.

C. Circuit Housing Compression Testing

Housing can withstand up to 4830N (equivalently 1000lb force) of compression force (measured using Mark10).

Final Prototype: Hunger Qs

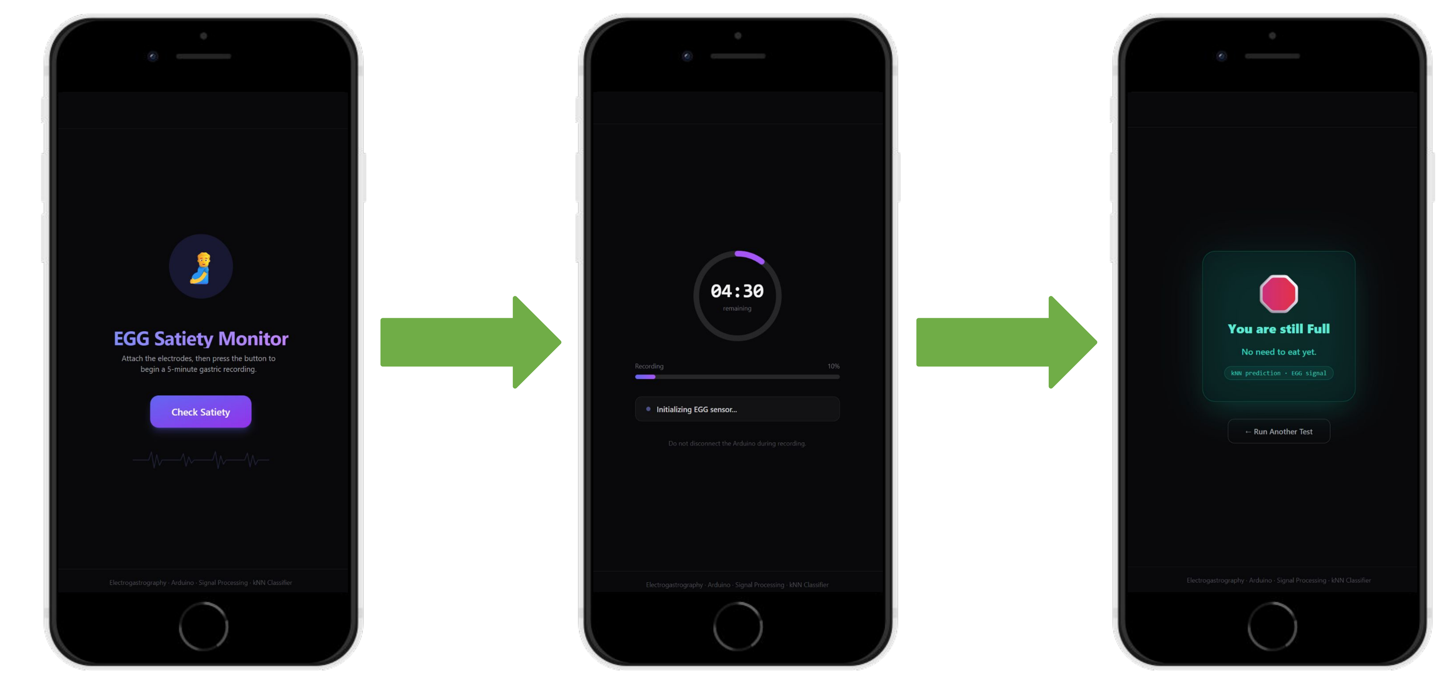


Figure 8. Application UI showing the user how to start the measurements, recording duration, and final results from EGG test.

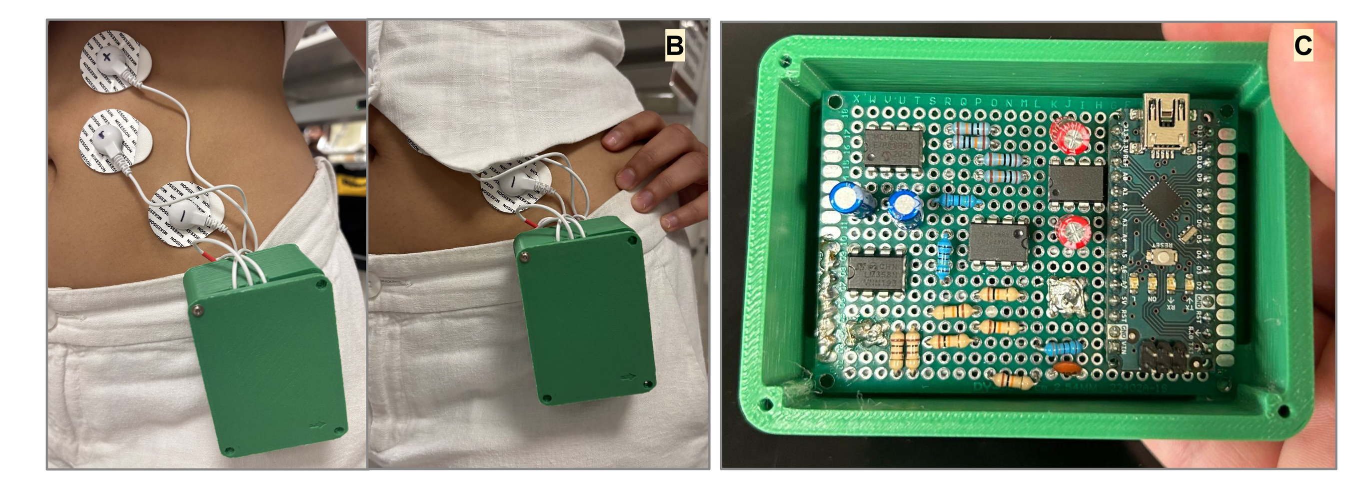


Figure 9. (A-B) EGG circuitry and electrodes clipped on model. **(C)** 3D printed housing containing perfboard soldered EGG circuitry.

- A clip-on wearable housing was engineered for a 5 minute EGG signal capture → transmitted to the Hunger Qs mobile app → classified and reports the user's hunger state

Conclusions

- Butterworth circuit, Welch analysis, and prediction website work effectively and cohesively for acquising and analyzing on EGG signals
- Hunger was predicted with an average accuracy of 77% across 100 different iterations of kNN models

Future Direction

- Sampling individuals using glucagon-like peptide 1 (GLP-1s) such as Ozempic®, shown to be a physiological regulator of appetite, to better understand the physiological effects for this emerging class of drugs [3]

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References

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[2] Ochoa, A.; Herrera-Lozada, J. C.; Cruz-Ortiz, D. *2025 11th International Conference on Control, Decision and Information Technologies (CoDIT) 2025*, 1472–1477.
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